

Conceptual Modelling of a Healthcare Process: Event-Based

System Overview:

This model represents patient flow through three stages: preparation, surgery in the operating theatre (OR), and recovery. Patients arrive according to a specified inter-arrival time distribution. Preparation facilities and recovery beds have limited capacity, while the OR is a single resource. The model focuses on measuring the key metrics:

- Average patient throughput time (arrival to discharge)
- Probability and duration of Operating theatre blocking time due to shortage of recovery beds.

Essential System Variables:

Variable	Description
Clock	Simulation time
prep_capacity	Number of preparation facilities
busy_prep	Number of preparation facilities currently in use
prep_queue	Queue of patients waiting for preparation
ot_busy	1 if operating theatre is occupied, 0 otherwise
surgery_queue	Queue of patients waiting for surgery
rec_capacity	Number of recovery beds
busy_rec	Number of recovery beds in use
recovery_queue	Patients waiting for recovery when no bed available (represents blocking state)
OR_blocked	Boolean: TRUE if operating theatre cannot release patient
blocking_start_time	Timestamp when OR blocking period begins
cumulative_blocking_time	Total accumulated blocking duration
arrival_time[p]	Arrival time per patient (for throughput calc)
cumulative_throughput_time	Sum of all patients' throughput times
num_completed_patients	Count of discharged patients

Event List and Actions:

Event 1: PATIENT ARRIVAL

Trigger: Scheduled based on inter-arrival distribution

Actions:

- Create new patient; record $\text{arrival_time}[p] = \text{Clock}$
- If $\text{busy_prep} < \text{prep_capacity}$:
 - Trigger PREPARATION_START for this patient
- Else:
 - Add patient to prep_queue
- Schedule next PATIENT_ARRIVAL

Event 2: PREPARATION START (helper event, conceptual only)

(Occurs whenever a patient moves from queue to preparation)

Trigger: A patient is assigned to begin preparation.

Actions:

- $\text{busy_prep} += 1$
- Schedule END_PREPARATION at $\text{Clock} + \text{prep_duration}$

Event 3: END PREPARATION

Trigger: Completion of preparation

Actions:

- busy_prep -= 1
- If prep_queue not empty → PREPARATION_START(next patient)
- If ot_busy == 0 and OR_blocked == FALSE:
 - Allocate OR → schedule END_SURGERY
- Else:
 - Add patient to surgery_queue

Event 4: OPERATION START (optional named event for clarity)

Actions:

- ot_busy = 1
- Schedule END_SURGERY at Clock + surgery_duration

Event 5: END SURGERY

Trigger: Completion of surgery

Actions:

- If busy_rec < rec_capacity:
 - busy_rec += 1
 - ot_busy = 0
 - Schedule END_RECOVERY
 - If surgery_queue not empty → OPERATION_START(next patient)
- Else:
 - OR_blocked = TRUE
 - blocking_start_time = Clock
 - Record patient as waiting for recovery (but patient stays inside OR)
 - The operating theatre cannot start the next surgery

Event 6: RECOVERY START (helper event, conceptual only)

Used when a bed becomes available after blocking

Actions:

- busy_rec += 1
- ot_busy = 0
- OR_blocked = FALSE
- Add blocking time: cumulative_blocking_time += (Clock - blocking_start_time)
- Schedule END_RECOVERY

Event 7: END RECOVERY

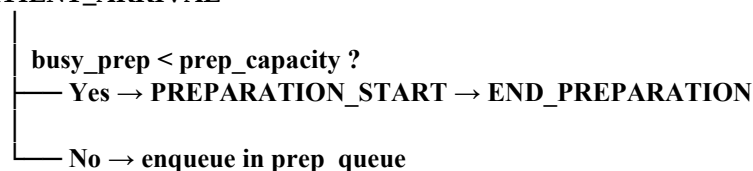
Trigger: Completion of recovery

Actions:

- busy_rec -= 1
- throughput_time = Clock - arrival_time[p]
- cumulative_throughput_time += throughput_time
- num_completed_patients += 1
- If recovery_queue not empty:
 - Remove next patient from recovery_queue
 - **Trigger RECOVERY_START for that patient**
 - If surgery_queue not empty → OPERATION_START(next patient)

Event Interactions Diagram

PATIENT_ARRIVAL



(next arrival event is always scheduled)

PREPARATION_START

— Schedule END_PREPARATION

END_PREPARATION

busy_prep -= 1

if prep_queue not empty → PREPARATION_START(next patient)

Is operating theatre free AND not blocked?

— Yes → OPERATION_START → END_SURGERY

— No → enqueue in surgery_queue

OPERATION_START

— Schedule END_SURGERY

END_SURGERY

busy_rec < rec_capacity ?

— Yes →

busy_rec += 1

ot_busy = 0

→ END_RECOVERY

If surgery_queue not empty → OPERATION_START(next patient)

— No →

OR_blocked = TRUE

blocking_start_time = Clock

Patient waits in recovery_queue

(patient remains occupying OR → ot_busy stays 1)

END_RECOVERY

busy_rec -= 1

throughput_time = Clock - arrival_time[p]

cumulative_throughput_time += throughput_time

num_completed_patients += 1

recovery_queue not empty ?

— Yes →

RECOVERY_START(next blocked patient)

If surgery_queue not empty → OPERATION_START(next patient)

— No → patient departs (end of lifecycle)

RECOVERY_START (after a blocked patient gets a freed bed)

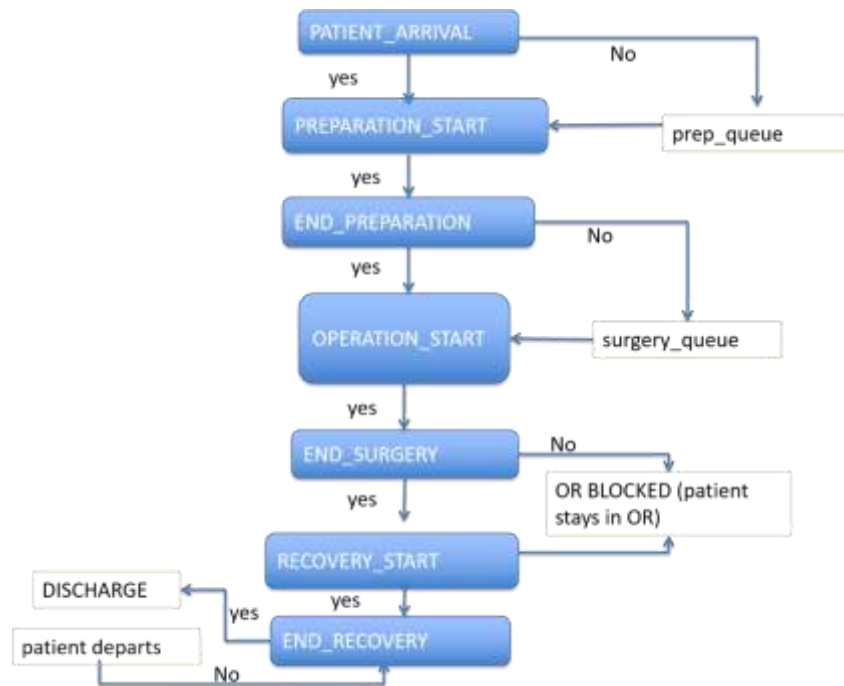
busy_rec += 1

ot_busy = 0

OR_blocked = FALSE

cumulative_blocking_time += (Clock - blocking_start_time)

— Schedule END_RECOVERY



Data Monitoring for Metrics

Throughput Time:

- At END_RECOVERY: $\text{throughput} = \text{Clock} - \text{arrival_time}[p]$
- $\text{Average throughput} = \text{cumulative_throughput_time} / \text{num_completed_patients}$

Blocking Probability:

- Blocking duration added each time RECOVERY_START occurs
- $\text{blocking_probability} = \text{cumulative_blocking_time} / \text{Clock}$

Event-Based Advantages:

- **Explicit control** over all state changes
- **Easier to model complex synchronization** (like OT blocking)
- **Better performance** for large-scale simulations
- **Clear causality** in event sequencing

Conclusion of Event-Based Model:

The event-based conceptual model correctly represents patient flow through limited preparation facilities, the single operating theatre, and limited recovery beds. The model explicitly captures the possibility of operating theatre blocking when recovery capacity is full by keeping the completed surgical patient inside the OR until a recovery bed becomes available. The monitoring of throughput time and blocking time is built directly into event transitions, ensuring the key performance measures are obtained.

Conceptual Modelling of a Healthcare Process: Process-Based

System Overview:

This model outlines a simulation of a patient stream through an operating theatre, addressing the required metrics: average patient throughput time and the operating theatre blocking time due to limited recovery facilities. The model is designed through several stages:

- Preparation: Patients arrive and are served by several identical preparation facilities.
- Surgery: Patients move to a single Operating Theatre (OT).
- Recovery: Operated patients move to a limited recovery facilities.

The essential system components and variables:

Components	Description	Process Type	Variables
Patients	The individual entities that flow through the system. Each patient represents an active "process."	Active Process	Arrival Time, Preparation Start/End Time, Surgery Start/End Time, Recovery Start/End Time, Departure Time, Individual Throughput Time.
Preparation Facilities	Shared resources. Patients request one when available.	Passive Process	Number of facilities acquired by patients.
Operating Theatre (OT)	A single resource. Patients request it for surgery.	Passive Process	Status (Acquired/Released), Time held by a patient while waiting for recovery (OT Blocking Time).
Recovery Facilities	Shared resources. Patients request one after surgery.	Passive Process	Number of facilities acquired by patients.

Decision on Active Processes:

Based on the movement and interactions, the primary active component in this simulation will be the Patient.

Patient (Active Process): Each patient entity will execute its own lifecycle, moving through the stages, requesting and releasing resources as needed. This allows for a clear representation of an individual patient's journey and how they interact with the system's resources. Each patient process is responsible for requesting resources (Preparation, Operating Theatre (OT), and Recovery), delaying for service, and tracking its own flow time.

Passive resources (Preparation Facilities, Operating Theatre, Recovery Facilities) will be managed by the simulation framework, which handles:

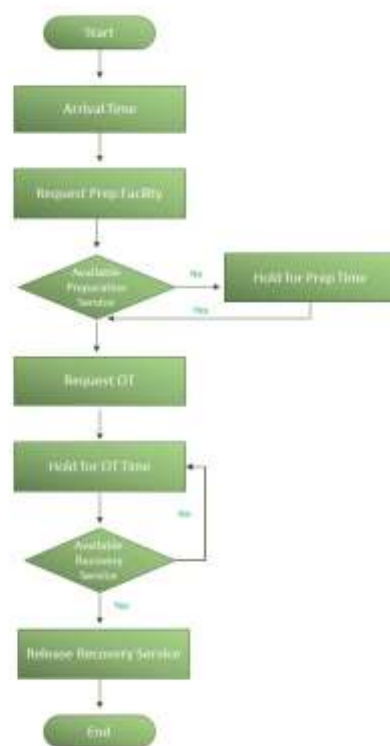
- Resource Allocation: When an active patient process calls request(resource), the framework checks if the resource is free.
- Queue Management: If the resource is busy, the framework automatically places the patient process into a queue associated with that resource. The patient process is then suspended until its turn comes.
- Automatic Unblocking: When a resource is released, the framework automatically resumes the next waiting patient process in the queue.

Patient Process Life Cycle Sketch:

1. **ARRIVE:**
 - Record Arrival Time.
 - Schedule next Patient Arrival (for Continuous Flow).
2. **WAIT FOR PREP:**

- REQUEST one Preparation Facility (from the pool).
 - Once acquired: Record Prep Start Time.
- 3. PREPARATION SERVICE:**
- WAIT for Prep Service Time (delay).
- 4. RELEASE PREP:**
- RELEASE Preparation Facility.
- 5. WAIT FOR SURGERY:**
- REQUEST the Operating Theatre (single resource).
 - Once acquired: Record Surgery Start Time.
- 6. SURGERY SERVICE:**
- WAIT for Surgery Service Time (delay).
- 7. WAIT FOR RECOVERY (CRITICAL BLOCKING POINT):**
- REQUEST one Recovery Facility (from the pool).
 - Once acquired:
 - RELEASE Operating Theatre.
 - Record Recovery Start Time.
 - IF BLOCKING OCCURS: The OT is held by the patient (not released at step 6) until a Recovery Facility becomes free. The time spent holding the OT while waiting for Recovery is the OT Blocked Time.
- 8. RECOVERY SERVICE:**
- WAIT for Recovery Service Time (delay).
- 9. DEPART:**
- RELEASE Recovery Facility.
 - Record Departure Time.
 - Calculate and record Throughput Time.

Flowchart Diagram:



Data Monitoring for Metrics (Process-Based Model)

Throughput Time:

Metric	Calculation Point	Logic
Individual Patient Throughput	At the final step: Patient Departs (after releasing the Recovery Facility).	$\text{throughput} = \text{Departure_Time} - \text{Arrival_Time}$
Average Throughput	Calculated at the end of the simulation based on aggregated patient data.	$\text{average_throughput} = \frac{\text{Sum of all individual throughput times}}{\text{Total number of patients who departed}}$

Blocking Probability:

Metric	Calculation Point	Logic
OT Blocking Duration	Tracked during the BLOCKING state (Step 6 in the life cycle).	When the patient process is suspended after surgery but before acquiring Recovery, measure the duration between the time the patient attempts to request Recovery and the time the request succeeds (i.e., when the process resumes). This duration is the blocking_time added to the cumulative total.
Blocking Probability	Calculated at the end of the simulation.	$\text{blocking_probability} = \frac{\text{Cumulative blocking time recorded by all patients}}{\text{Total simulated clock time (end time)}}$

Process-Based Advantages:

- **Natural Flow Representation:** The model follows the **narrative** of the patient's journey, making the code intuitive and highly readable.
- **Simplified Resource Handling:** The simulation framework automatically manages queues and resource availability, abstracting away complex synchronization logic for resources (like Preparation Facilities).
- **Direct Blocking Measurement:** The critical blocking time is measured naturally by having the patient process hold one resource (OT) while waiting for another (Recovery).
- **Ease of Extension:** Adding new sequential stages or conditional decisions to the patient's flow is often simpler as just need to insert a step into the patient's existing lifecycle.